

Database Technologies-Assignment 6

Use Group by clause with appropriate SQL functions to solve following queries.

Note: To solve below queries, use “sales” database

1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).

```
KD2_98846_Raj>SELECT odate, COUNT(DISTINCT snum) AS number_of_salespeople FROM ORDERS GROUP BY odate;
```

odate	number_of_salespeople
1990-10-03	4
1990-10-04	3

2 rows in set (0.01 sec)

2. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form: For the city (city), the highest rating is: (rating).

```
KD2_98846_Raj>SELECT CONCAT('For the city ',city,'the highest rating is: ',MAX(rating)) AS result FROM customers GROUP BY city;
```

result
For the city Londonthe highest rating is: 100
For the city Romethe highest rating is: 200
For the city San Josethe highest rating is: 300
For the city Berlinthe highest rating is: 300

4 rows in set (0.00 sec)

3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

```
KD2_98846_Raj>SELECT odate, SUM(amt) AS total_orders  
-> FROM orders  
-> GROUP BY odate  
-> ORDER BY total_orders DESC;
```

odate	total_orders
1990-10-04	16713.81
1990-10-03	8944.59

2 rows in set (0.00 sec)

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table. (Note Use the average amount value directly →2565.84)

```
KD2_98846_Raj>SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > 2565.84;
```

```
+-----+-----+
| snum | total_amount |
+-----+-----+
| 1001 |      15382.07 |
| 1002 |       5546.15 |
+-----+-----+
2 rows in set (0.01 sec)
```

5. Write a query that selects the highest rating in each city.

```
KD2_98846_Raj>SELECT city, MAX(rating) AS highest_rating
-> FROM customers
-> GROUP BY city;
```

```
+-----+-----+
| city      | highest_rating |
+-----+-----+
| London    |             100 |
| Rome      |             200 |
| San Jose  |             300 |
| Berlin    |             300 |
+-----+-----+
4 rows in set (0.00 sec)
```

6. Retrieve the maximum order amount for each salesperson from the orders table, including only those orders whose value exceeds Rs. 3000.

```
KD2_98846_Raj>SELECT snum, MAX(amt)
-> FROM orders
-> WHERE amt > 3000
-> GROUP BY snum;
```

snum	MAX(amt)
1002	5160.45
1001	9891.88

2 rows in set (0.00 sec)

7. Write an SQL query to find the smallest order placed by every customer.

```
KD2_98846_Raj>SELECT cnum, MIN(amt)
-> FROM orders
-> GROUP BY cnum;
```

cnum	MIN(amt)
2008	18.69
2001	767.19
2007	1900.10
2003	5160.45
2002	1713.23
2004	75.75
2006	4723.00

7 rows in set (0.00 sec)